

Operator Discovery Report: stagnation_aware_acceptance_balanced_restart

- Operator type: acceptance.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministically controls bounded-worse acceptance during local search and cools later, to escape 2opt plateaus without excessive deterioration; tuned for EUC_2D transfer and two-cluster bottlenecks.
- Novelty classification: threshold_acceptance.
- Rediscovery signature: acceptance:threshold:thr=0.018:temp=0.005.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Use threshold acceptance when a restart solution is not immediately improving.
- Relax or tighten acceptance depending on stagnation and late-search cooling.
- Reject candidate restarts that exceed the configured worse-move budget.

Complexity

- Non-empty lines: 11.
- Complexity score: 0.42.
- Fingerprint: d8743608d5a41495.

Why It Should Help

- The operator only changes restart acceptance, so any gain is attributable to a narrow exploration-vs-stability decision rule.

Failure Cases

- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 28.551443, distance-eval delta 0.0.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 1588.360657, distance-eval delta 0.0.
- random_restart_2opt gap delta 0.0, runtime delta -69.4037, distance-eval delta -2310.285714.
- sparse_three_opt gap delta 0.0, runtime delta -7.085743, distance-eval delta 0.0.
- clustered_local_search gap delta 0.008268, runtime delta 17.945586, distance-eval delta 2075.428571.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.0.
- Runtime inflation: -0.001558.
- Distance-evaluation inflation: 0.0.
- Same-gap-faster flag: True.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.0.
- improving_only_acceptance mean gap 0.121317 and delta vs full operator 0.0.
- tight_threshold_acceptance mean gap 0.121317 and delta vs full operator 0.0.